

Presented by Elliot Feng, Elizabeth Chiou,
Jerry Yang, and Elaine Zhang (Group 10)

Predicting Hit Songs

from Spotify Audio Features

Which combination of song features best predicts whether a song is a hit? We define a hit as a song with a Spotify popularity score ≥ 70 .

Data & Setup

I Don't Care (with Justin Bieber) - Loud Luxury Remix	Ed Sheeran	68	3uCuX0GfAR086GHALE2DCL	I Don't Care (with Justin Bieber) [Loud Luxury Remix]	2019-06-14	Pop Remix
Memories - Dillon Francis Remix	Maroon 5	67	63P5G26AU6WTKX56vYv6	Memories (Dillon Francis Remix)	2019-12-13	Pop Remix
All the Time - Don Diablo Remix	Zara Larsson	70	1Hs5m2kLcUrR0vE8y7Yv4	All the Time (Don Diablo Remix)	2019-07-05	Pop Remix
Call You Mine - Keanu Silva Remix	The Chainsmokers	60	1nqYv0eTfYKXuGOVh0v48	Call You Mine - The Remixes	2019-07-19	Pop Remix
Someone You Loved - Future Humans Remix	Lewis Capaldi	69	7e7v9vHqG40LfUuRZ2vQ	Someone You Loved (Future Humans Remix)	2019-03-05	Pop Remix
Beautiful People (feat. Khalid) - Jack Wins Remix	Ed Sheeran	67	2y9vR0QdNvNCE0U0N	Beautiful People (feat. Khalid) (Jack Wins Remix)	2019-07-11	Pop Remix
Never Really Over - K3HAB Remix	Katy Perry	62	78vY7SeuafYmH0vDnm8pE	Never Really Over (K3HAB Remix)	2019-07-26	Pop Remix
Post Malone (feat. RAN) - GATTUSO Remix	Sam Feldt	69	670358PULS4BPMFYJas1	Post Malone (feat. RAN) (GATTUSO Remix)	2019-08-29	Pop Remix
Tough Love - Tiesto Remix / Radio Edit	Avicii	68	7CuAR0vq48HwE8T9b0aw	Tough Love (Tiesto Remix)	2019-06-14	Pop Remix
If I Can't Have You - Gryffin Remix	Shawn Mendes	67	4Qu0T5vnyEQuwFFEV5u	If I Can't Have You (Gryffin Remix)	2019-06-20	Pop Remix
Cross Me (feat. Chance the Rapper & PhB Rock) - M-22 Remix	Ed Sheeran	58	48wK28KvG0uQvG5Cv0a8v0	Cross Me (feat. Chance the Rapper & PhB Rock) (M-22 Remix)	2019-06-21	Pop Remix
Hide Me - K3HAB Remix	Ellie Goulding	67	6e6fYFV62wTFLAZ0CMpa	Hide Me (K3HAB Remix)	2019-08-16	Pop Remix
Body On My SOS - Laidback Luke Tribute Remix / Radio Edit	Loud Luxury	67	28660ncp3huuL0r0WQ5P	Body On My SOS (Laidback Luke Tribute Remix)	2019-03-29	Pop Remix
Summer Days (feat. Macklemore & Patrick Stump of Fall Out Boy) - Tiesto Remix	Avicii	68	7u4T0dC5wNLS0nYfu	Summer Days (feat. Macklemore & Patrick Stump of Fall Out Boy) (Tiesto Remix)	2019-05-17	Pop Remix
South of the Border (feat. Camila Cabello & Cardi B) - Andy Jarvis Remix	Martin Garrix	63	2v4388MhE43ygl0T0Z1	Summer Days (feat. Macklemore & Patrick Stump of Fall Out Boy) (Remixes)	2019-07-12	Pop Remix
All My Friends - Eden Prince Remix	Ed Sheeran	66	1n0kAvL0wvP7uFM7JPF9	South of the Border (feat. Camila Cabello & Cardi B) (Andy Jarvis Remix)	2019-12-11	Pop Remix
Say My Name (feat. Bebe Rexha & J Balvin) - Lucas & Steve Remix	AJ Mitchell	60	6288B6l9vTc0aqpLXV1	All My Friends (Eden Prince Remix)	2019-05-03	Pop Remix
Dancing With A Stranger (With Normani) - Cheat Codes Remix	David Guetta	65	19c3v68T89Z54w0CFL1	Say My Name (feat. Bebe Rexha & J Balvin) (Lucas & Steve Remix)	2019-12-28	Pop Remix
Let It Be Me - SonDr5 Remix	Sam Smith	69	2LDwN4Z4q5SA3Z9C04Q	Dancing With A Stranger (With Normani) [Cheat Codes Remix]	2019-03-08	Pop Remix
Heaven - David Guetta & MORTEN Remix	Steve Aoki	35	097Yycr5ZuLS0PwvUhat	Let It Be Me (Remixes)	2019-10-18	Pop Remix
All Around The World (La La La) - Marik Remix	Avicii	70	37cTCDGLWMy0cBwNKAxB	Heaven (David Guetta & MORTEN Remix)	2019-08-23	Pop Remix
Don't Leave Me Alone (feat. Anne-Marie) - EDE's Indian Summer Remix	K3HAB	64	5v858UJ8MvY0ZT7YvQwT	All Around The World (La La La) (Marik Remix)	2019-08-23	Pop Remix
Slow Dance (feat. Ava Max) - Sam Feldt Remix	David Guetta	62	78NFMFM0vATJ0v8h3CSQ	Don't Leave Me Alone (feat. Anne-Marie) (EDE's Indian Summer Remix)	2019-05-08	Pop Remix
Do You Mean (feat. Ty Dolla Sign & Jélie) - Myon Remix	AJ Mitchell	70	60MC4Z0T8wvTul8v8hWv	Slow Dance (feat. Ava Max) (Sam Feldt Remix)	2019-10-25	Pop Remix
Hold Me While You Wait - Steve Void Remix	The Chainsmokers	58	5vT68vRX82f3vS5kxUJT	Do You Mean - The Remixes (feat. Ty Dolla Sign & Jélie)	2019-05-24	Pop Remix
One Thing Right - Finesse Remix	Lewis Capaldi	63	1LTVUJQ9vNvRvbm0Z20vL	Hold Me While You Wait (Steve Void Remix)	2019-08-21	Pop Remix
Close To Me (with Diplo) (feat. Swae Lee) - CID Remix	Marshmello	66	13v0v0D0vppvUvka4wG	One Thing Right (Remixes)	2019-08-09	Pop Remix
So Am I - Toby Green Remix	Ellie Goulding	8	58AMFMvN28ZNT0u8h3uA	Close To Me (Remixes)	2019-02-01	Pop Remix
When You Leave - Breathe Carolina Remix	Ava Max	62	4p2838FY2YpqT7YwUuA	So Am I (Toby Green Remix)	2019-05-23	Pop Remix
Mother - Fedde Le Grand Remix	Niki Valina	30	6M86WEAP8Xvq0TtU0c	When You Leave (Breathe Carolina Remix)	2019-04-19	Pop Remix
Signs - Eden Prince Remix	Charlie Puth	56	28yv2XGPTvWvGM84qgo	Mother (Fedde Le Grand Remix)	2019-11-01	Pop Remix
Run My Life - Steve James Remix	Storley	56	24wAA8M85304vUw4k8E	Signs (Eden Prince Remix)	2019-11-30	Pop Remix
Tom - KREAM Remix	Zara Larsson	58	6v8H9yLcUc4G08vC85LN	Run My Life (Remixes)	2018-11-23	Pop Remix
Quite Miss Home - Steve Void Remix	Ava Max	60	6pKLNK8P80405vyp4MD	Tom (KREAM Remix)	2019-09-05	Pop Remix
Love Me Less (feat. Quinn XCII) - Party Pupils Remix	James Arthur	60	4H87x1uRZC5UAT7vqPEK	Quite Miss Home (Steve Void Remix)	2019-11-22	Pop Remix
Side Effects - Fedde Le Grand Remix	MAX	55	3E38AaZ280v0w0PwuvWv	Love Me Less (Remixes)	2019-06-21	Pop Remix
Good Things Fall Apart (with Jon Bellion) (Tiesto's Big Room Remix)	Marshmello	59	44w6E2yv0H2vCZv8KJ	Happier (Remixes Pt. 2)	2019-11-30	Pop Remix
Paradise - Bazzi vs. Black Caviar Remix	The Chainsmokers	63	37v039vJh8v0uPhy0t0at	Side Effects - Remixes	2019-07-25	Pop Remix
	ILLENIUM	67	1v4KEHvGJvJh36Fv8v2v8e	Good Things Fall Apart (with Jon Bellion) (Tiesto's Big Room Remix)	2019-07-12	Pop Remix
	Bazzi vs.	55	0TLVJLJ0Gv157UUKvAMx	Paradise (Bazzi vs. Black Caviar Remix)	2019-06-27	Pop Remix

25,736 unique tracks

10.2% of the songs in the dataset are hits

24 subgenres across 6 genres

~32,000 songs before feature engineering

We used artist-level split, a method to divide the dataset such that artist's songs appear exclusively in the training or test sets, to prevent the model from cheating by recognizing a specific artist's songs in the test set.

EDA

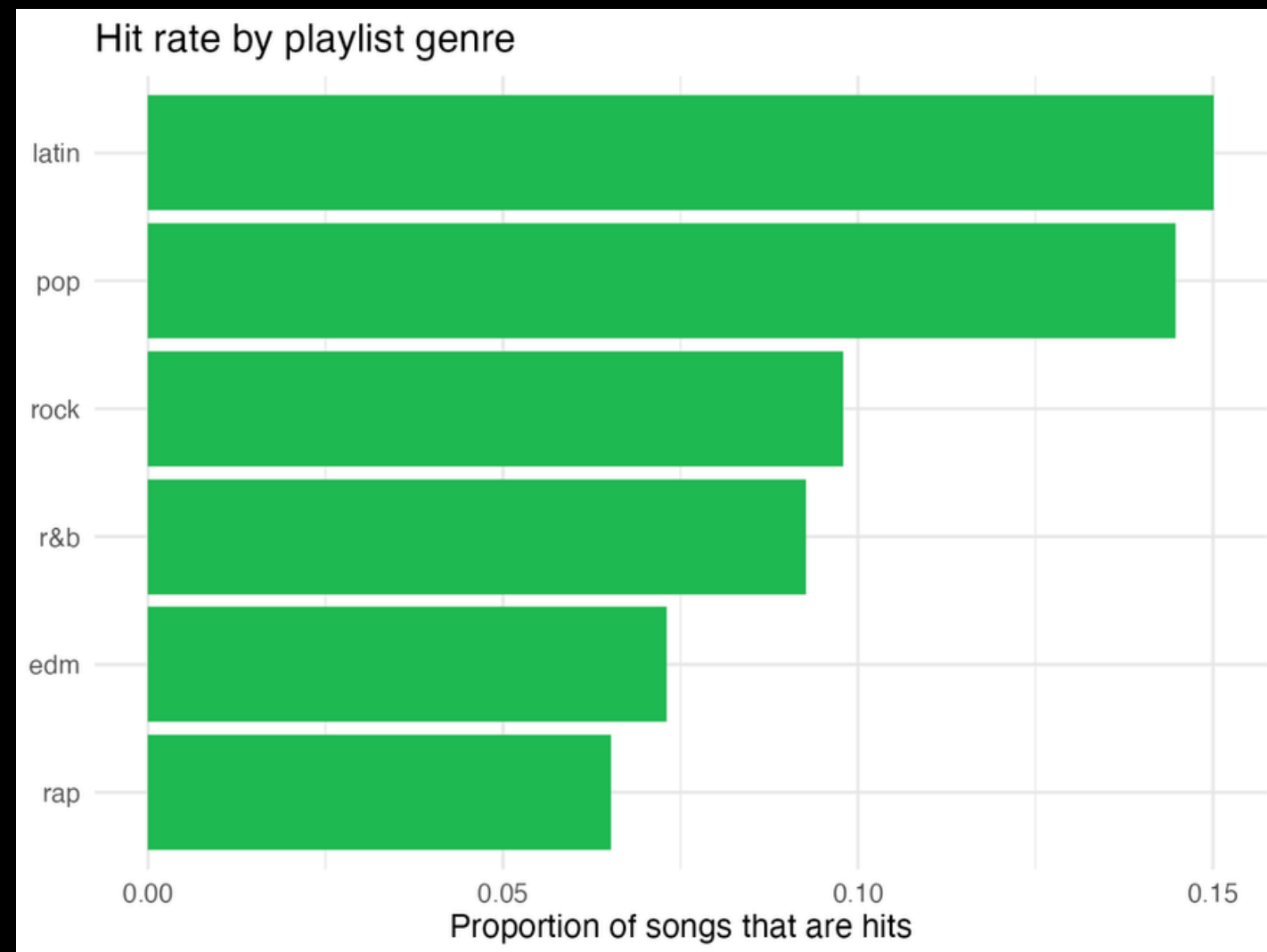


Figure 1. Hit rate across 6 broad genres. Data: full cleaned dataset (n = 25,736).

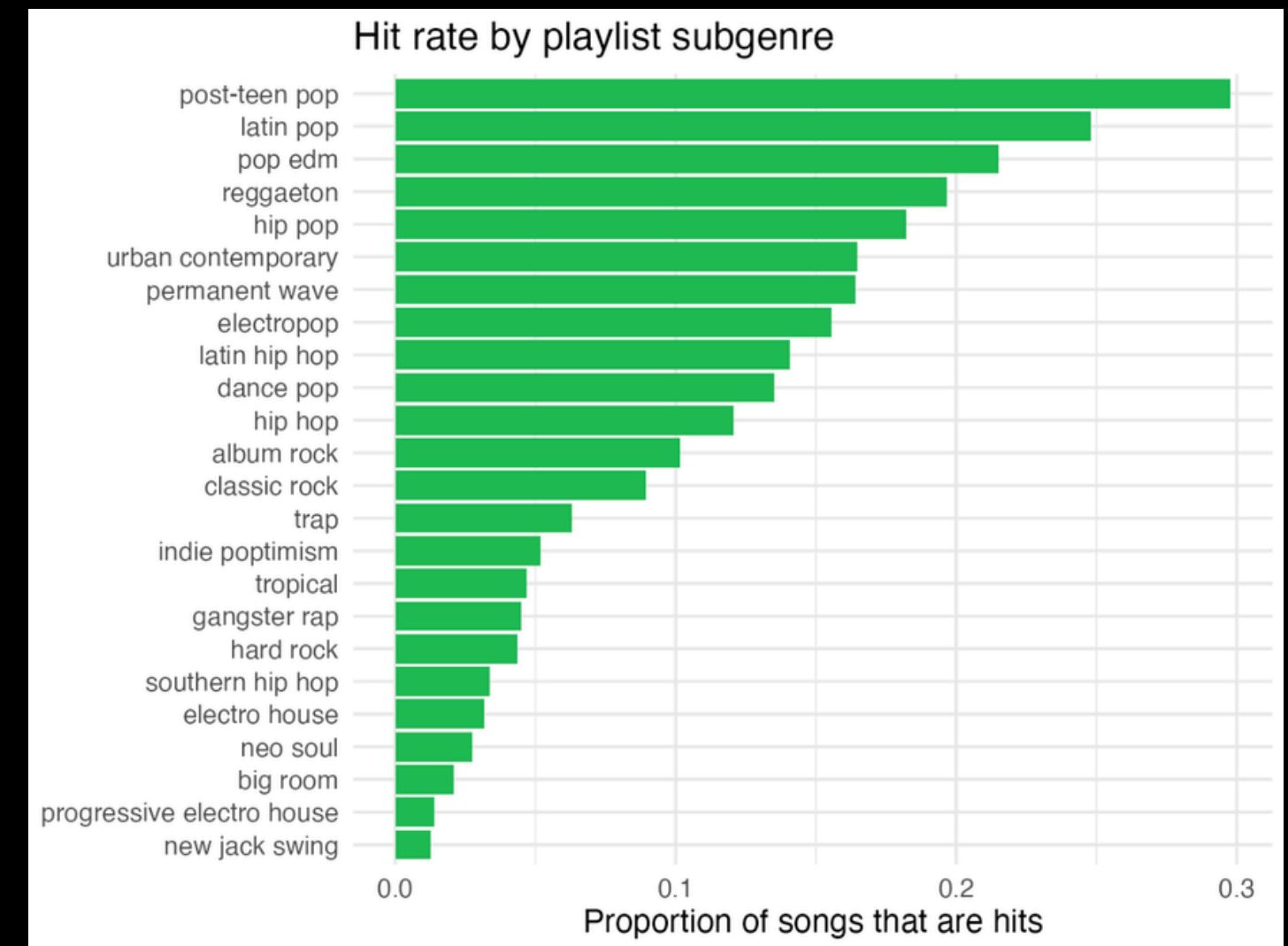


Figure 2. Hit rate across 24 playlist subgenres. Data: full cleaned dataset (n = 25,736).

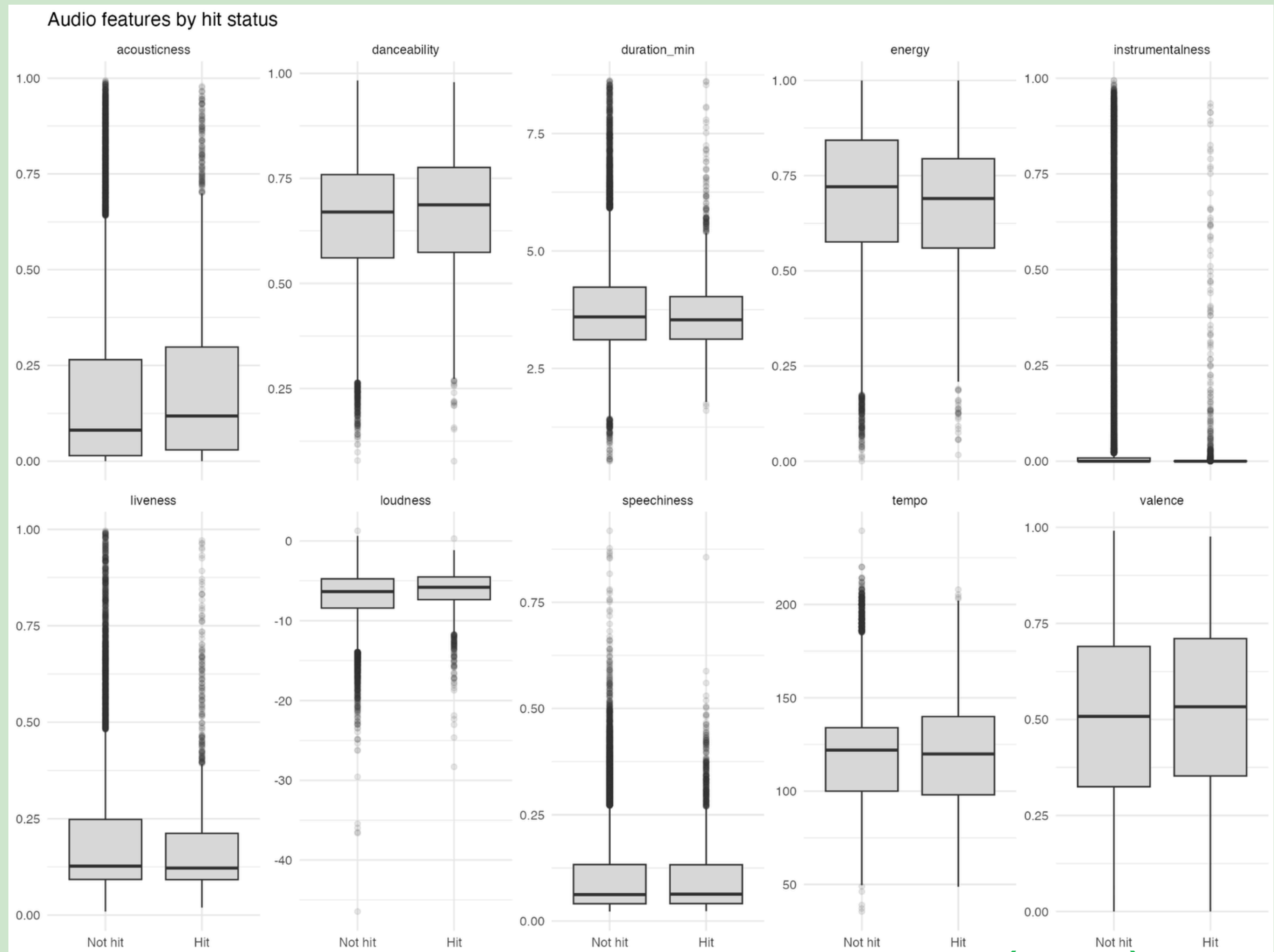


Figure 3. Audio feature distributions by hit status. Data: full cleaned dataset (n = 25,736).

EDA

Model Roadmap

Nested
logistic
models

MODEL 1

Audio only

Baseline using
Spotify audio
features.

MODEL 2

Audio + genre

Adds the six broad
playlist genres.

MODEL 3

Full logistic model

Uses subgenre, song
age, and mode instead
of broad genre.

MODEL 4

Backward selection

Starts from the full
model and removes
weak predictors by AIC.

Independent
checks

CHECK 1

LASSO

Logistic regression with a
cross-validated penalty.

CHECK 2

Random forest

Tree ensemble benchmark for
flexible feature interactions.

CHECK 3

Neural network

Small nonlinear benchmark
with standardized numeric
inputs.

Same judge for every model: performance on a held-out test set of artists the models never saw during training.

Logistic Regression Model

To prevent overfitting we used:

- Ten-fold cross validation
- Folds grouped by artist
- **0.77 $\pm$ 0.03**

At our cutoff, the model catches ~72% of true hits while correctly clearing 67% of non-hits.

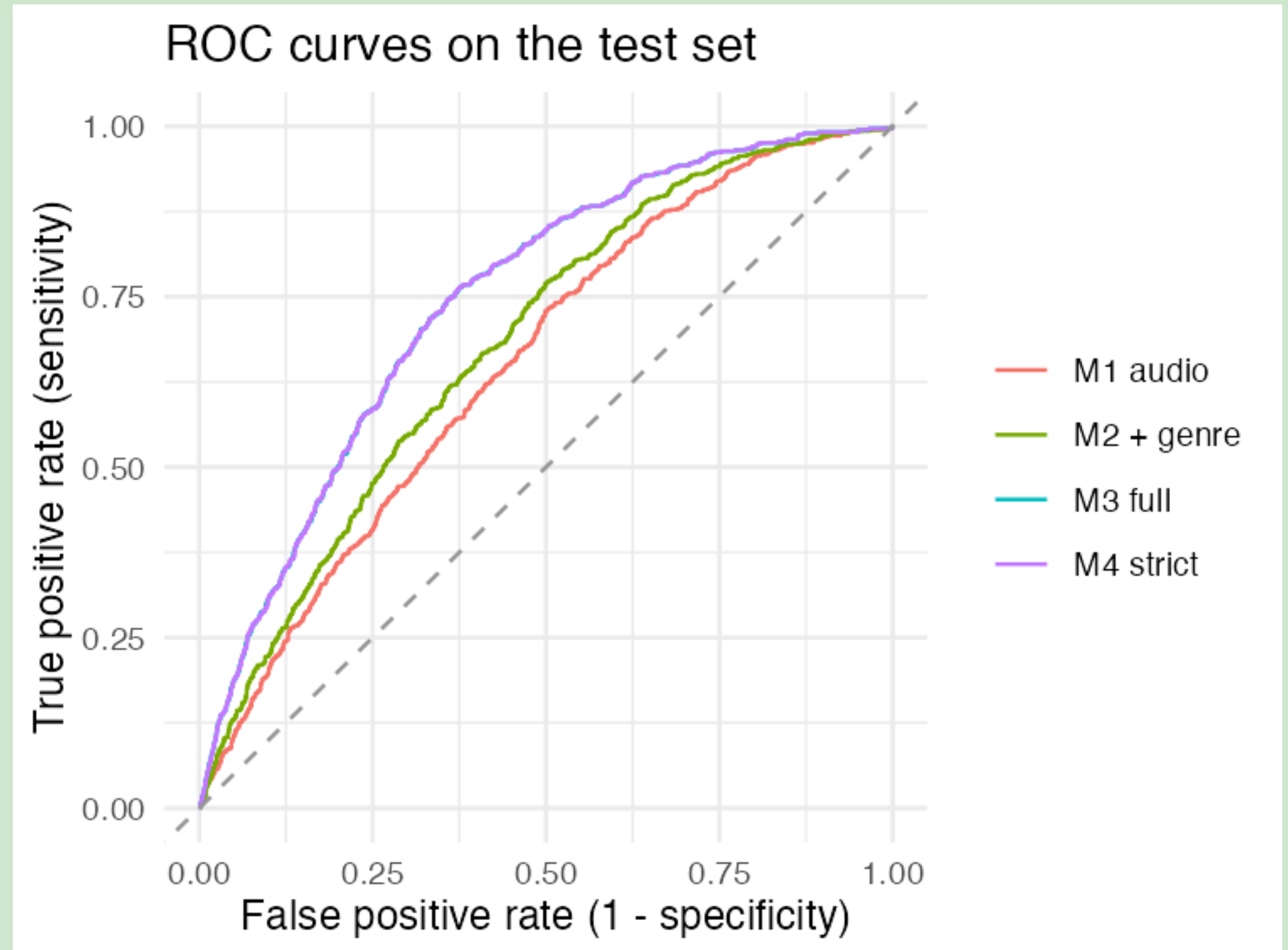


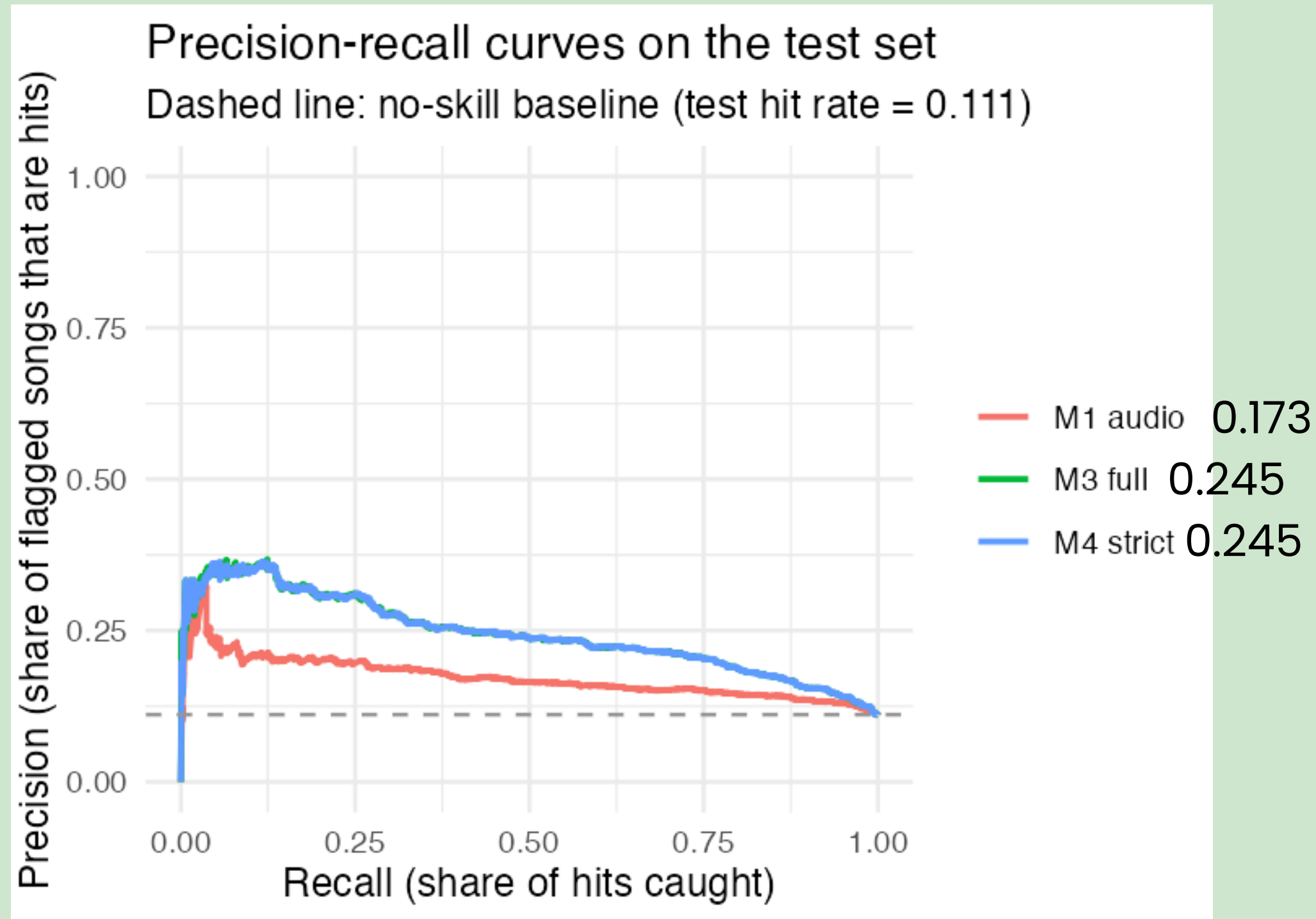
Figure 4. ROC curves for the four logistic models. Data: held-out test set (n = 5,155).

Accuracy

Model	Terms	AUC	Accuracy	F1	Sensitivity	Precision	Specificity
M1 audio	10	0.651	0.567	0.250	0.650	0.155	0.556
M2 + genre	11	0.681	0.612	0.270	0.647	0.171	0.607
M3 full	13	0.744	0.672	0.327	0.719	0.212	0.666
M4 strict (best)	8	0.743	0.669	0.323	0.710	0.209	0.664

Table 1. Test-set performance of the four logistic models (held-out test data, n = 5,155). Bold = best value in each column; green row = chosen model.

Precision–Recall



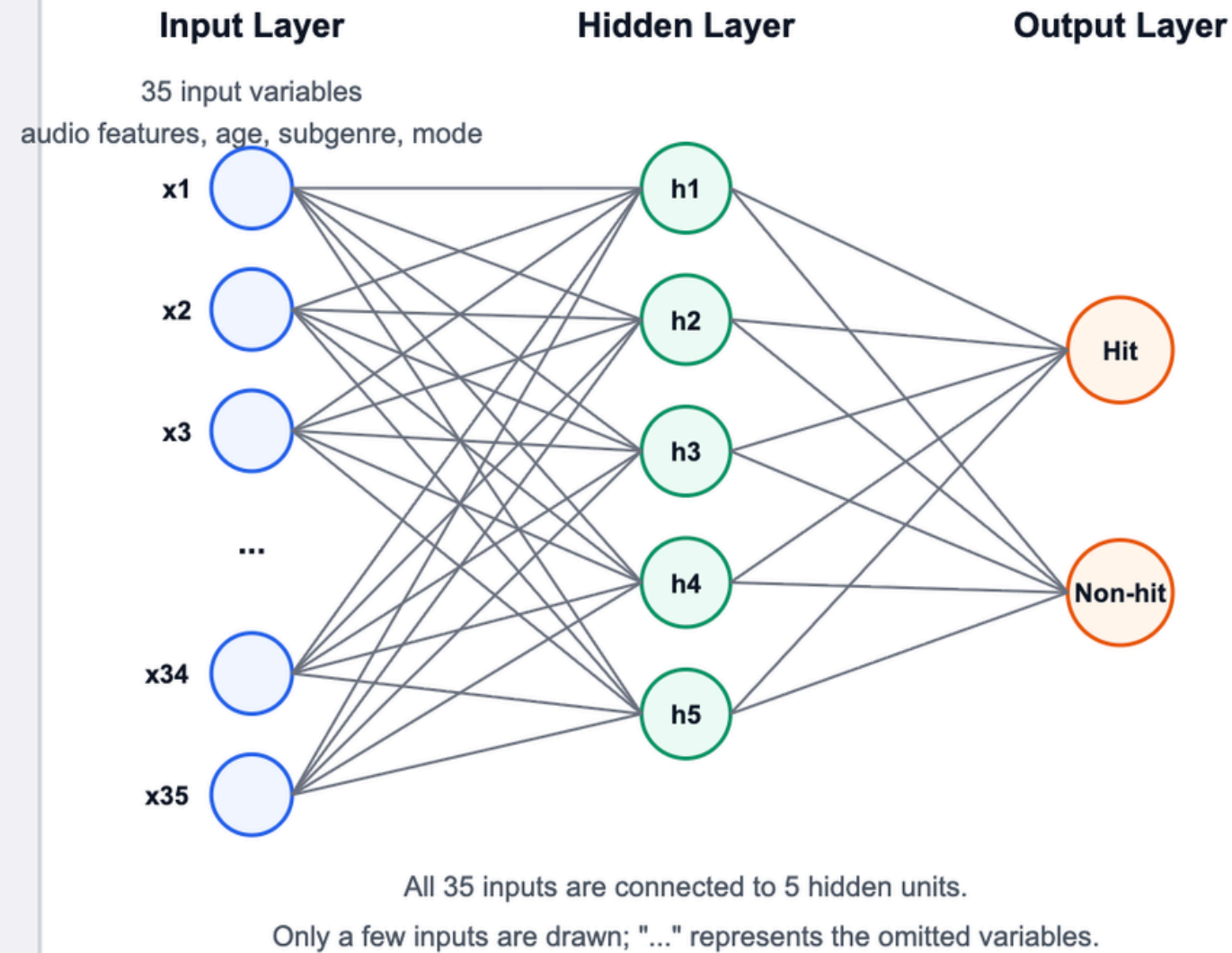
Our best model's average precision is 0.245, against a no-skill baseline of 0.111 which is more than twice as good as chance.

Figure 5. Precision–recall curves. Data: held-out test set (n = 5,155).

Neural Network and Random Forest

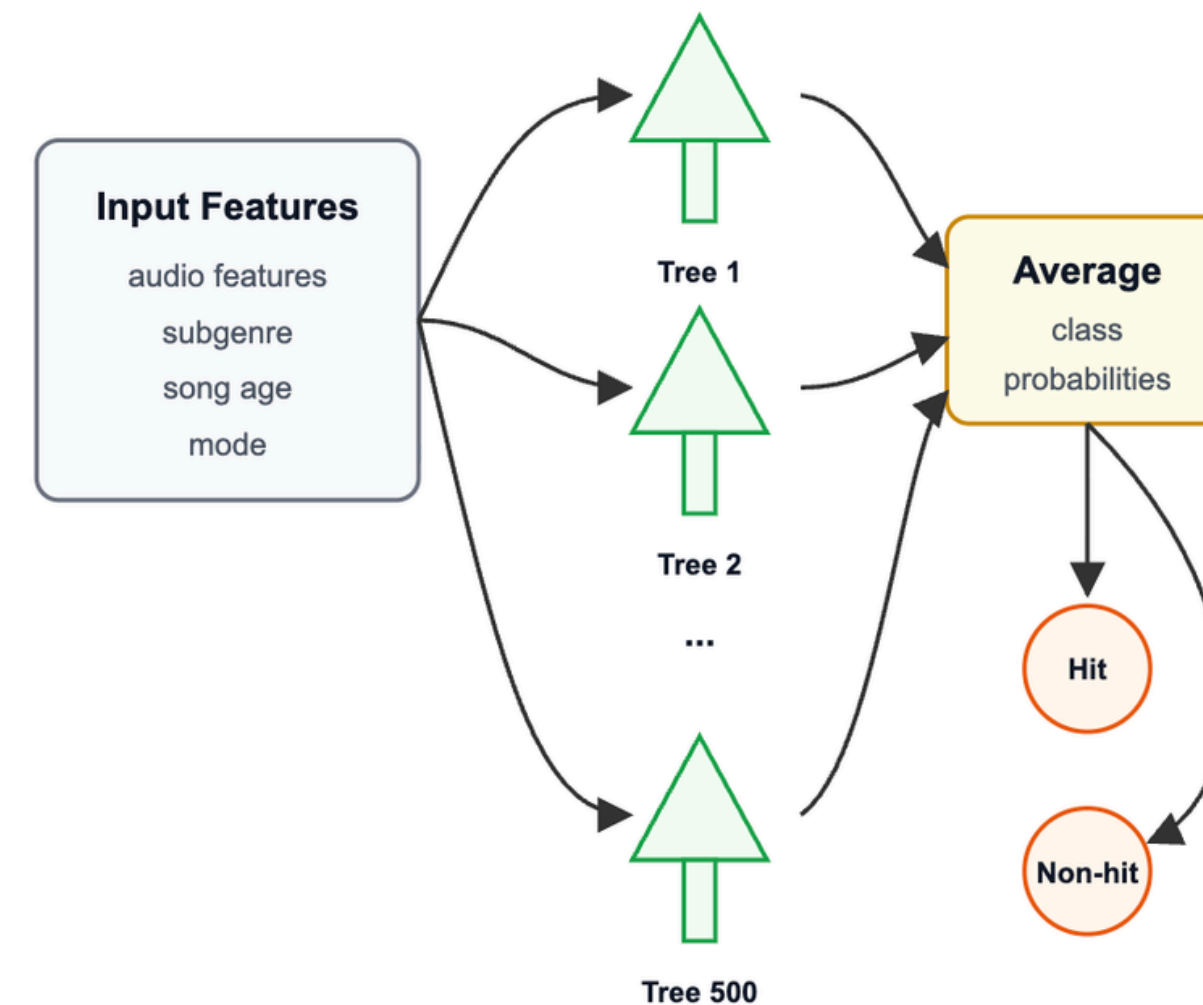
One-Hidden-Layer Neural Network

Classification model with 5 hidden units



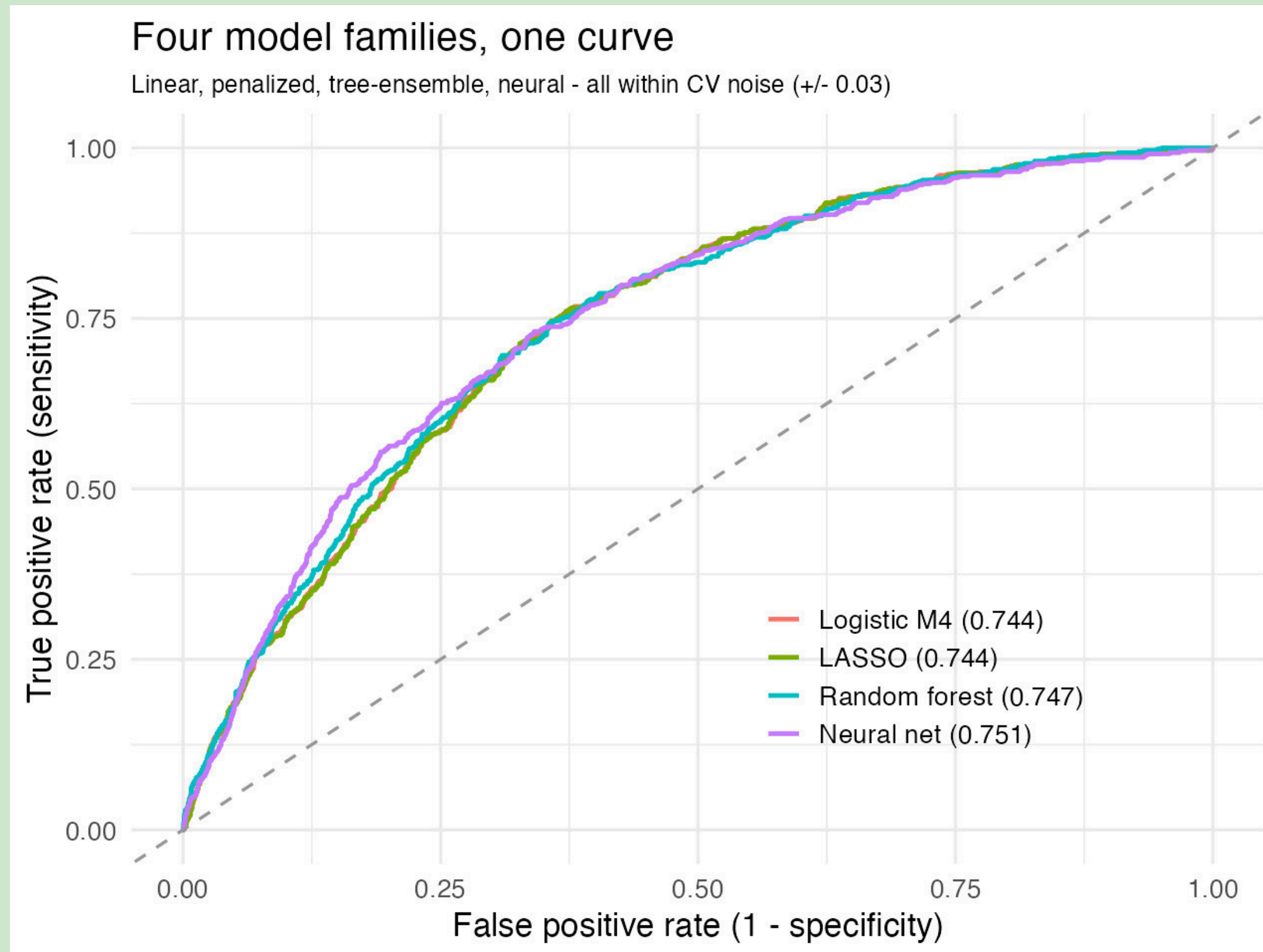
Random Forest Classifier

500 decision trees with averaged class probabilities



Each tree produces class probabilities; the forest averages them for the final prediction.

The 4 Different Models



Four unrelated model families, one curve.

Logistic: 0.744. LASSO: 0.744. RF: 0.747. NN: 0.751. All gaps sit inside CV noise (± 0.03).

The bottleneck isn't the model. It's the features. The ceiling is around 0.75, and it belongs to the data.

Figure 6. ROC curves for the four model families. Data: held-out test set ($n = 5,155$).

What Predicts a Hit

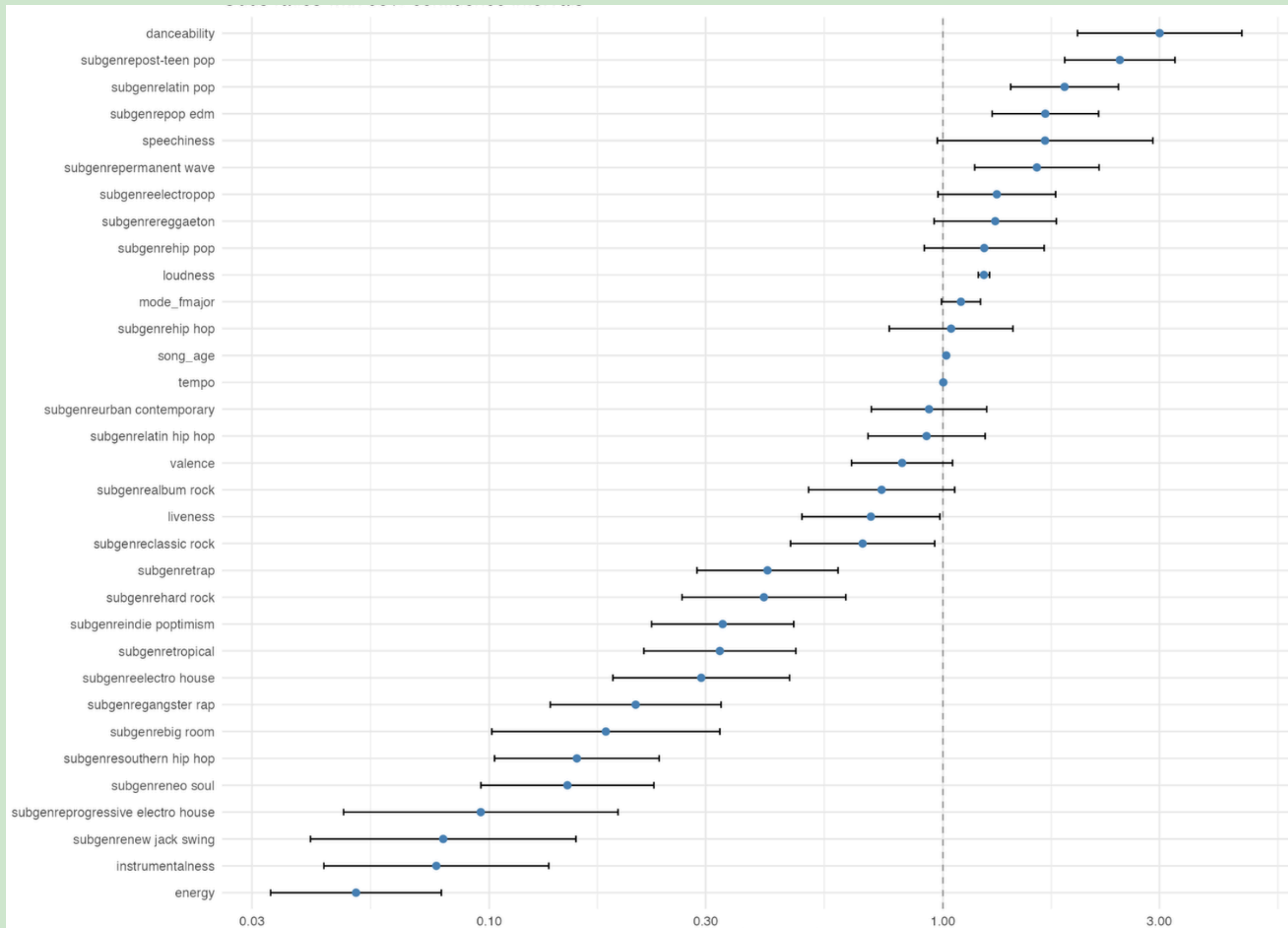


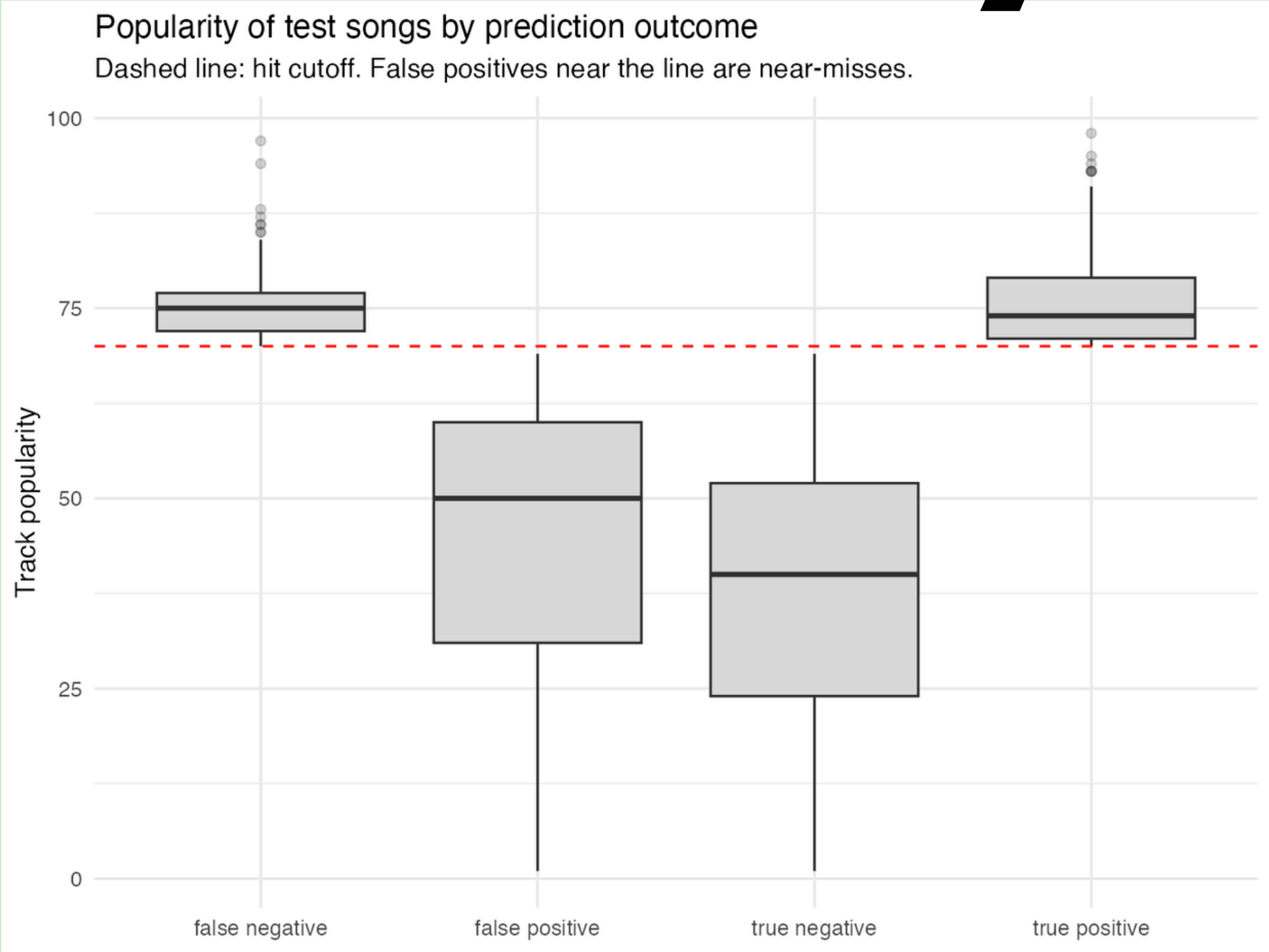
Figure 7. Odds ratios with 95% CIs from the best model (M4). Model fit on training data (n = 20,581).

Anova: Each Variable Carries Real Signal

Predictor	LR χ^2	df	p-value
Subgenre	1,173.6	23	< 0.001
Loudness	216.9	1	< 0.001
Energy	210.5	1	< 0.001
Instrumentalness	134.3	1	< 0.001
Song age	28.1	1	< 0.001
Danceability	26.6	1	< 0.001
Tempo	4.9	1	0.027
Liveness	4.1	1	0.042

Table 2. Type-II likelihood-ratio tests for the predictors in M4. Model fit on training data (n = 20,581).

Error Analysis



		Actual	
Predicted	0	1	
	0	3054	161
1	1529	411	

Table 3. Confusion matrix at the hit-rate cutoff. Data: held-out test set (n = 5,155).

Missed hits: median ~75 — just above the cutoff.

False alarms: median ~50 — well above typical non-hits at 40.

Figure 8. Popularity of songs by prediction outcome (best model, M4). Data: held-out test set (n = 5,155).

Leakage

Artist-only leakage model

AUC = 0.873

Best audio-based model

AUC = 0.744

Metric	Leakage model
AUC	0.873
Accuracy	0.845
F1	0.548
Sensitivity	0.846
Precision	0.405
Specificity	0.845

Table 4. Artist-only leakage model vs best audio model. Data: held-out test set (n = 5,155).

Model Comparison

Model	AUC	Accuracy	F1	Sensitivity	Precision	Specificity
M1 audio	0.651	0.567	0.250	0.650	0.155	0.556
M2 + genre	0.681	0.612	0.270	0.647	0.171	0.607
M3 full	0.744	0.672	0.327	0.719	0.212	0.666
M4 strict (chosen model)	0.743	0.669	0.322	0.710	0.209	0.664
LASSO min	0.744	0.672	0.326	0.717	0.211	0.666
LASSO lse	0.734	0.667	0.312	0.682	0.202	0.665
Neural net	0.751	0.702	0.332	0.666	0.221	0.707
Random forest	0.747	0.632	0.315	0.760	0.198	0.616

Table 5. Performance of all model families. Data: held-out test set (n = 5,155). Bold = best value in each column; green row = chosen model (M4).

Conclusion



Audio and subgenre get you to an AUC of about 0.75.
Four different model families all agree that is the ceiling.



Loudness is the strongest signal. Subgenre gives you
the biggest lift.



A song's success comes more from the artist's reputation,
marketing, and playlist placement than from the audio itself.



The biggest limitations: popularity measured at a single point in time,
so cause and effect cannot be separated. The sample comes from
curated playlists, not a random draw of all music. Songs that never got
playlisted are not included.

AI Acknowledgment

We acknowledge the assistance of AI tools used during the completion of this project. We utilized AI to debug, create graphs, and finish repetitive tasks. AI was used exclusively as a learning and editing aid, resolving challenges throughout the research process.

Works Cited

Yamaç Eren Ay. "Spotify Dataset 1921-2020, 600k+ Tracks." Kaggle, www.kaggle.com/datasets/yamaerenay/spotify-dataset-19212020-600k-tracks. Accessed 9 July 2026.

Presented by Elliot Feng, Elizabeth Chiou,
Jerry Yang, and Elaine Zhang (Group 10)

Predicting Hit Songs

from Spotify Audio Features

Which combination of song features best predicts whether a song is a hit? We define a hit as a song with a Spotify popularity score ≥ 70 .